HOFA: Channel-Wise Hybrid Attention via Outlier Factorization

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Abstract

KV cache memory remains a primary bottleneck during long-context autoregressive decoding. While prior hybrid architectures mitigate this by alternating exact and linear attention across macro structural axes (depth, width, or sequence), they impose a coarse tradeoff between decoding efficiency and fine-grained associative recall. We introduce Hybrid Outlier-Factorized Attention (HOFA), a channel-wise intra-head decomposition. Inside every attention head, HOFA statically allocates a small r-dimensional outlier subspace to exact causal Softmax attention while routing the remaining inlier channels to a recurrent Gated Linear Attention (GLA) state. Hardware profiling demonstrates that this intra-head split reduces the decoding KV-cache footprint by 43.7% and yields up to a +75.9% peak decoding throughput speedup over standard MHA. Pretraining across 70M, 125M, and 350M parameter scales establishes that HOFA matches MHA perplexity and achieves competitive zero-shot reasoning accuracy. We show that this factorization naturally concentrates global associative matching into the outlier channels while the linear recurrence absorbs background context, with retrieval limits closely following a proposed theoretical extreme-value capacity model.

https://github.com/LotanGutman/OutlierNeuronAttention

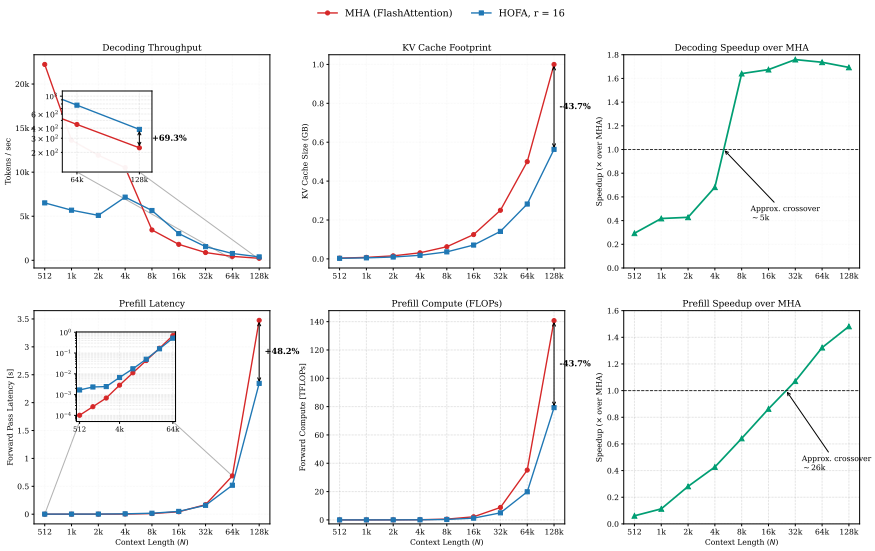


Figure 1: **Hardware Execution & Throughput Benchmarks.** Empirical autoregressive decoding throughput and KV-cache footprint scaling (top row) alongside prefill latency and compute scaling (bottom row) comparing HOFA to FlashAttention-2 MHA.

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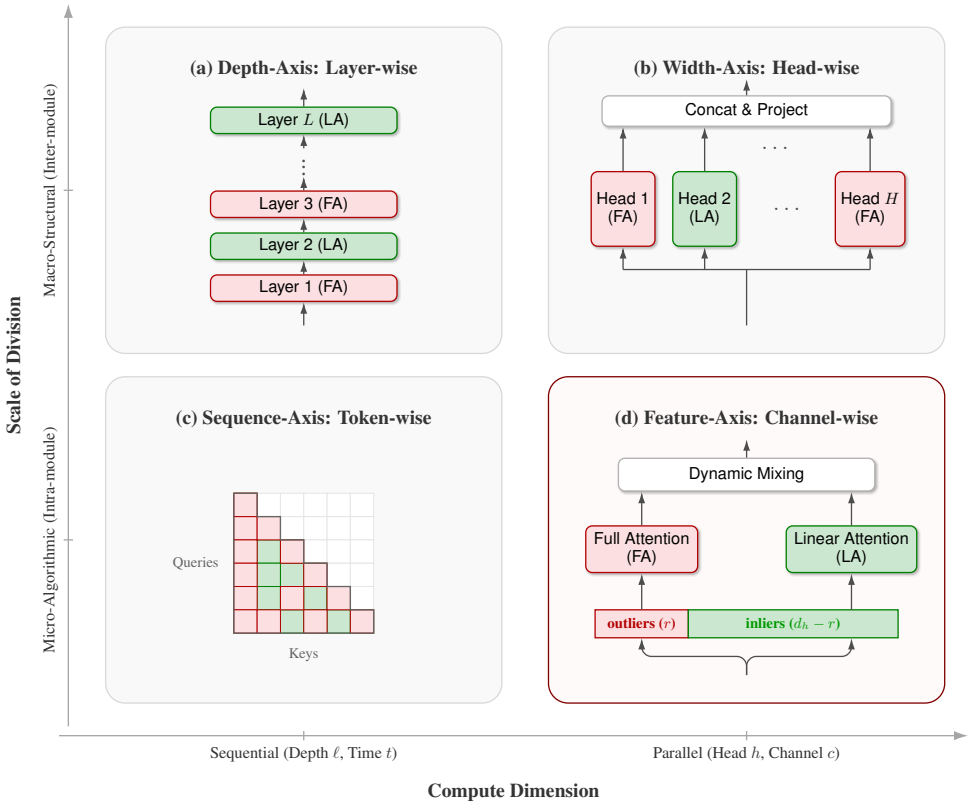


Figure 2: **The Architectural Taxonomy of Hybrid Attention.** Existing hybrid models decompose the attention operation along the (a) layer, (b) head, or (c) token dimension. HOFA introduces the (d) feature dimension axis, processing every token simultaneously through disjoint exact (**Softmax**) and linear (**GLA**) subspaces within the same head.

1 Introduction & Related Work

Multi-Head Attention (MHA) [Vaswani et al., 2017] forms the computational core of modern Large Language Models (LLMs), but decoding is memory-bandwidth bottlenecked over long contexts. For a sequence length N, standard MHA must cache d_h-dimensional Key and Value vectors for every token across all heads and layers, incurring an $\mathcal{O}(NLHd_h)$ memory footprint per sequence. During generation ($N = 1$), memory bandwidth is dominated by streaming this large-scale KV-cache from High-Bandwidth Memory (HBM) into SRAM, limiting decoding throughput [Dao et al., 2023].

To address this memory bottleneck, recent research has explored hybrid architectures that combine exact $\mathcal{O}(N^2)$ Softmax attention with $\mathcal{O}(1)$ linear-time State-Space Models (SSMs) [Gu and Dao, 2023, Dao and Gu, 2024] or linear attention mechanisms [Peng et al., 2023, Yang et al., 2023, 2025]. Existing hybrid approaches decompose attention along three macro structural axes (Figure 2): (i) **depth-axis hybrids** alternating pure Softmax and linear layers sequentially across depth (e.g., StripedHyena [Poli et al., 2023], Jamba [AI21 Labs, 2024]); (ii) **width-axis hybrids** assigning distinct full-attention and linear-attention heads in parallel within each layer (e.g., Hymba [Dong et al., 2024]); and (iii) **sequence-axis hybrids** dynamically routing token positions or sliding-window blocks to exact attention while processing background context linearly (e.g., Longformer [Beltagy et al., 2020], Native Sparse Attention [Yuan et al., 2025]).

While existing hybrids multiplex across depth, width, or sequence dimensions (Figure 2a-c), HOFA introduces the fourth architectural axis: channel-wise intra-head decomposition (Figure 2d). Inside every attention head, HOFA statically splits the hidden dimension d_h into a small r-dimensional outlier channel $(\widetilde{Q}_{\mathcal{O}}, \widetilde{K}_{\mathcal{O}})$ and a remaining $(d_h - r)$-dimensional inlier channel $(\widetilde{Q}_{\mathcal{I}}, \widetilde{K}_{\mathcal{I}})$. The r outlier dimensions receive exact causal Softmax attention with Rotary Position Embeddings (RoPE) [Su et al., 2024], while the inlier dimensions are processed by a data-dependent GLA state.

This architectural factorization is motivated by systematic feature outliers in trained Transformers [Dettmers et al., 2022, Pierro and Abreu, 2024, An et al., 2025]. Quantization studies [Dettmers et al., 2022, Pierro and Abreu, 2024] demonstrate that extreme activation magnitudes cluster within a few systematic feature dimensions, which route specific semantic sub-tasks [An et al., 2025]. Optimization concentrates discriminative factual matching into a small subset of outlier channels ($\widetilde{Q}_{\mathcal{O}}, \widetilde{K}_{\mathcal{O}}$), while offloading background context to a recurrent GLA inlier state. Here, "outlier" refers to the high-magnitude feature subspace learned by the model; the allocation is architectural rather than determined by a runtime outlier detector. By establishing a dedicated spatial channel (r) inside every attention head, HOFA allows these feature outliers to operate with exact causal Softmax attention while avoiding full KV-cache memory overheads.

We formalize our contributions across three core axes. First, we model HOFA's context capacity limits via the Welch bound, consistent with empirical induction sequence degradation. Second, hardware profiling demonstrates a 43.7% KV-cache footprint reduction and a +75.9% peak decoding throughput speedup over MHA. We motivate this architectural choice by analyzing attention mass distribution across production pretrained models. Finally, pretraining across 70M, 125M, and 350M scales demonstrates perplexity parity with MHA and strong zero-shot reasoning performance under extended $2\times$ Chinchilla-optimal budgets [Hoffmann et al., 2022].

2 Methodology & Theoretical Framework

2.1 Architectural Formulation

Figure 3 illustrates the HOFA token mixer architecture. Inside every attention head, representations are factorized into isolated exact and linear processing streams. Following linear projection, Query and Key representations are statically partitioned into an r-dimensional outlier subspace ($\widetilde{Q}_{\mathcal{O}}, \widetilde{K}_{\mathcal{O}}$) and a $(d_h - r)$-dimensional inlier subspace ($\widetilde{Q}_{\mathcal{I}}, \widetilde{K}_{\mathcal{I}}$). The outlier stream executes exact causal Softmax attention with Rotary Position Embeddings (RoPE) to isolate high-magnitude associative matching, while the inlier stream drives a Gated Linear Attention (GLA) recurrence stabilized by RMSNorm [Zhang and Sennrich, 2019] and scaled by a learnable channel-wise scale vector $\mathbf{s}_{\mathcal{I}}$ to absorb diffuse background context. A learned, context-aware gate α_h dynamically blends both pathway outputs before final projection $\mathbf{W}_O$.

Let $\mathbf{X} \in \mathbb{R}^{B\times N\times d_{\text{model}}}$ denote the input sequence. For each attention head $h \in \{1, \ldots, H\}$, query, key, and value representations are projected via linear weight matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d_{\text{model}}\times d_h}$:

$$\widetilde{Q} = \mathbf{X}\mathbf{W}_Q, \quad \widetilde{K} = \mathbf{X}\mathbf{W}_K, \quad V = \mathbf{X}\mathbf{W}_V \quad \in \mathbb{R}^{B\times N\times d_h} \tag{1}$$

Inside each head, we partition the channels into the first r outlier dimensions and the remaining $j = d_h - r$ inlier dimensions:

$$\widetilde{Q} = \left[\widetilde{Q}_{\mathcal{O}} \parallel \widetilde{Q}_{\mathcal{I}}\right], \quad \widetilde{K} = \left[\widetilde{K}_{\mathcal{O}} \parallel \widetilde{K}_{\mathcal{I}}\right] \tag{2}$$

For the outlier pathway, we apply RoPE exclusively to the r-dimensional subspace: $\widehat{Q}_{\mathcal{O}} = \text{RoPE}(\widetilde{Q}_{\mathcal{O}})$, $\widehat{K}_{\mathcal{O}} = \text{RoPE}(\widetilde{K}_{\mathcal{O}})$. The outlier output $Y_{\mathcal{O}}$ is computed using exact causal Softmax attention with variance normalization factor $S = \sqrt{d_h/r}$:

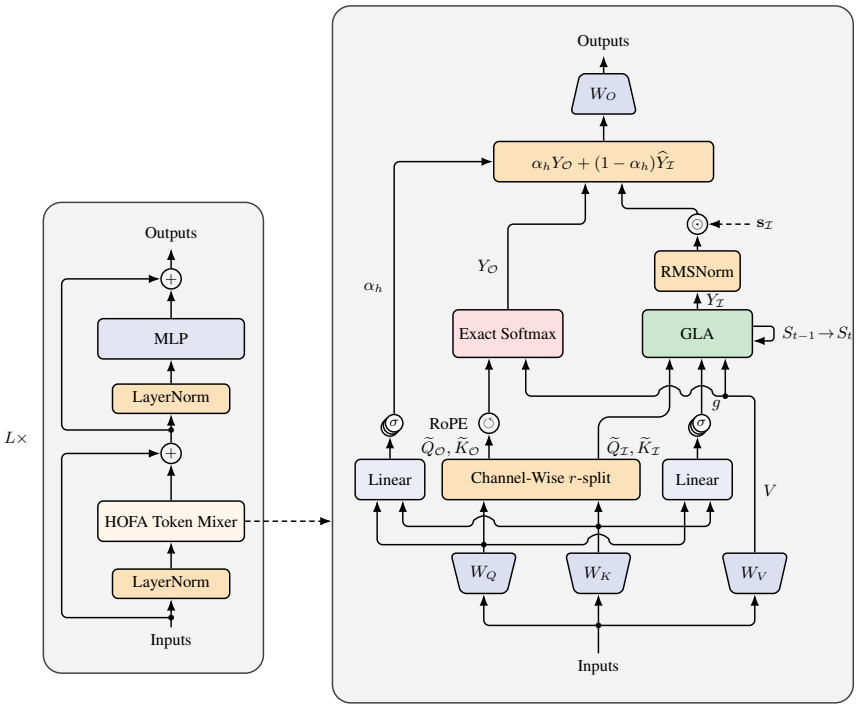
$$Y_{\mathcal{O}} = \text{softmax}\left(\frac{S}{\sqrt{d_h}}\widehat{Q}_{\mathcal{O}}\widehat{K}_{\mathcal{O}}^T + \mathbf{M}\right)V = \text{softmax}\left(\frac{1}{\sqrt{r}}\widehat{Q}_{\mathcal{O}}\widehat{K}_{\mathcal{O}}^T + \mathbf{M}\right)V \tag{3}$$

where $\mathbf{M}$ is the standard causal mask ($\mathbf{M}_{i,j} = 0$ for $i \geq j$, $-\infty$ otherwise).

For the inlier pathway, $\widetilde{Q}_{\mathcal{I}}$ and $\widetilde{K}_{\mathcal{I}}$ drive a GLA recurrence kernel with data-dependent decay $g_t = \sigma(\mathbf{W}_g[\widetilde{Q}_t, \widetilde{K}_t] + \mathbf{b}_g)$:

$$S_t = g_t \odot S_{t-1} + \widetilde{K}_{\mathcal{I},t}^T V_t, \quad Y_{\mathcal{I},t} = \widetilde{Q}_{\mathcal{I},t} S_t \tag{4}$$

The inlier output is normalized via Root Mean Square Normalization (RMSNorm) and scaled by a learnable channel-wise inlier scale vector $\mathbf{s}_{\mathcal{I}} \in \mathbb{R}^{d_h}$ (initialized to 1):



(a) Hybrid HOFA Transformer **(b) HOFA Token Mixer (intra-head)**

Figure 3: **Architecture of HOFA. (a)** Pre-LN Transformer layer containing the HOFA token mixer. **(b)** Inside the per-head mixer, projected queries and keys are statically split into r outlier channels (processed by exact causal Softmax attention) and remaining inlier channels (processed by Gated Linear Attention). Both pathways process the full value matrix V, with inlier outputs stabilized via RMSNorm and dynamically fused using a context-aware mixing gate α_h.

$$\widehat{Y}_\mathcal{I} = \mathbf{s}_\mathcal{I} \odot \mathrm{RMSNorm}(Y_\mathcal{I}) \tag{5}$$

Finally, both streams are dynamically blended via scalar gating $\alpha_h = \sigma(\mathbf{W}_\alpha[\widetilde{Q}, \widetilde{K}])$:

$$Y = \alpha_h \odot Y_\mathcal{O} + (1-\alpha_h) \odot \widehat{Y}_\mathcal{I}, \quad \text{Output} = Y\mathbf{W}_O \tag{6}$$

2.2 Computational Complexity & Limiting Regimes

For sequence length N, layer count L, head count H, and hidden dimension d_h, standard causal MHA evaluates $QK^\top$ and AV over $\frac{1}{2}N^2$ causal token pairs, requiring $\mathrm{FLOPs}_{\mathrm{MHA,attn}}(N) = 2LHd_hN^2$ FLOPs. In HOFA, exact causal Softmax evaluates $Q_\mathcal{O}K_\mathcal{O}^\top$ on the r-dimensional outlier subspace, and multiplies attention weights by the full V, yielding an attention-core cost of $LH(r+d_h)N^2$. Accounting for chunkwise inlier GLA recurrence ($\approx 6LH(d_h-r)d_hN$), total forward prefill FLOPs is:

$$\mathrm{FLOPs}_{\mathrm{HOFAPrefill}}(N) = 2PN + LH(r+d_h)N^2 + 6LH(d_h-r)d_hN \tag{7}$$

As $N \to \infty$, the quadratic term dominates, yielding the asymptotic attention-core prefill speedup ceiling relative to causal MHA:

$$\lim_{N\to\infty} \mathrm{Speedup}_{\mathrm{Prefill,attn}}(N) = \frac{2LHd_hN^2}{LH(r+d_h)N^2} = \frac{2d_h}{r+d_h} \tag{8}$$

For autoregressive decoding ($N=1$), memory bandwidth becomes the primary bottleneck. HOFA mitigates this by storing only r-dimensional keys ($K_\mathcal{O}$) alongside full Values (V) and a compact $j \times d_h$ linear state (S), resulting in the footprint ratio:

$$\lim_{N\to\infty} \frac{\mathrm{KV}_{\mathrm{HOFA}}}{\mathrm{KV}_{\mathrm{MHA}}} = \frac{r+d_h}{2d_h} \tag{9}$$

and asymptotic decoding speedup limit:

$$\lim_{N\to\infty} \text{Speedup}_{\text{Decode}}(N) = \frac{2d_h}{r + d_h} \tag{10}$$

For $d_h = 128$ and $r = 16$, Equation 9 predicts a footprint ratio of $\frac{16+128}{256} = 0.5625$ (-43.7% memory reduction) and Equation 10 yields a maximum theoretical decoding throughput speedup ceiling of $\frac{256}{144} \approx 1.777\times$ (+77.7%).

Limiting Architectural Regimes. The HOFA formulation exhibits smooth mathematical continuity between two canonical limiting regimes. In **Pure Linear Attention** ($r = 0$), the outlier subspace vanishes, reducing output to the normalized inlier stream $Y = \mathbf{s}_{\mathcal{I}} \odot \text{RMSNorm}(Y_{\mathcal{I}})$; the quadratic vanishes, collapsing to exact linear attention compute of $6LHd_h^2N$ FLOPs (or $(2P + 6LHd_h^2)N$ FLOPs for prefill). In **Dense Softmax Attention** ($r = d_h$), the inlier subspace vanishes, the linear term cancels to zero, and the quadratic term resolves to $LH(d_h + d_h)N^2 = 2LHd_hN^2$, recovering standard causal Multi-Head Attention FLOPs.

2.3 Capacity-Inspired Retrieval Model

We model HOFA's retrieval limits by framing the exact pathway as a continuous associative memory constrained by the aggregate interference floor of the Welch bound [Welch, 1974]. As derived in Appendix E, geometric cross-talk bounds the naive SNR ceiling by $\text{SNR}_{\text{naive}} \leq \frac{r}{cN-r}$, which isotropic scaling cannot overcome. Cross-entropy optimization drives an asymmetric gain factor γ within the r outlier channels against $n = N/d_h - 1$ competing distractors, yielding effective $\text{SNR}_{\text{eff}} \approx \gamma\frac{r}{cN-r}$ and logit mean $\mu = \sqrt{\text{SNR}_{\text{eff}}}$. Framing target retrieval as an M-ary hypothesis testing problem governed by Gumbel extreme value statistics [Fisher and Tippett, 1928] yields retrieval probability:

$$P(\text{correct}) \approx \Phi\left(\frac{\mu - b_n}{\sigma}\right) \tag{11}$$

where $b_n \approx \sqrt{2\ln(N/d_h - 1)}$ is the leading-order Gumbel threshold and σ is an empirical calibration parameter. Extended proofs and theoretical derivations are detailed in Appendix E.

3 Experimental Results

3.1 Hardware Execution & Efficiency

Experimental Settings. Appendix A (Table 2) details model configurations and full pretraining hyperparameters. All models are trained using PyTorch bfloat16 mixed-precision and AdamW (WD $= 0.1$) with Cosine Annealing. Guided directly by our empirical Feature-Wise Activation Norm Decomposition analysis across model depth (Section 3.2), we adopt a heterogeneous routing schedule. Concentrating outlier capacity ($r = 64, 32$) at boundary layers captures uncompressed lexical embeddings and next-token logit projections, while the middle $\sim$ 50–70% semantic core tapers to $r = 16$. Full layer-by-layer r-schedules for 70M ($L = 8$), 125M ($L = 12$), and 350M ($L = 24$) models are detailed in Appendix A (Table 3).

Computational Efficiency & Hardware Profiling. We benchmark HOFA against FlashAttention-2 MHA up to $N = 128$k on an NVIDIA RTX 4060 GPU using PyTorch mixed-precision (bfloat16) under dimensions $d_{\text{model}} = 2048, H = 16, d_h = 128, r = 16$. To isolate the core attention mechanism, we evaluate a single attention layer block with identity projections. During autoregressive decoding ($N = 1$), HOFA achieves a 43.7% KV-cache reduction and sustains a +69.3% throughput speedup over MHA at 128k context length, reaching a +75.9% peak speedup at 32k tokens (Figure 1, top), coming close to the $1.77\times$ theoretical bound (Equation 10). During prefill ($N > 1$), HOFA yields a -43.7% forward compute (FLOPs) reduction and a +48.2% latency speedup at $N = 128$k tokens, a gap that widens further as N scales toward the quadratic regime (Figure 1, bottom).

3.2 Mechanistic Validation & Capacity Limits

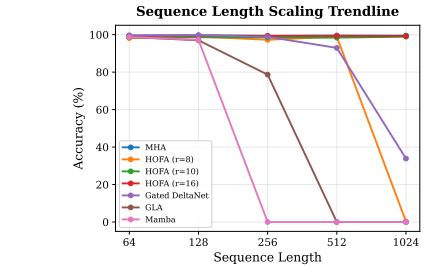
The Associative Recall Hierarchy. To evaluate sequence-scaling limits on associative recall, we benchmark 4-layer models ($L = 4, d_{\text{model}} = 128, H = 4, d_h = 32$) comparing Dense MHA,

HOFA ($r = 8, 10, 16$), Gated DeltaNet, GLA, and Mamba-2 on synthetic Induction Head retrieval [Olsson et al., 2022] across context windows $N \in [640, 1200]$. As shown in Figure 4a, the evaluated architectures fall into four empirical performance regimes, which we refer to as an Associative Recall Hierarchy: Class I (Exact Softmax: MHA, HOFA $r \geq 10$) maintains full retrieval accuracy up to $N = 1024$; Class II (Dynamic Matrix Gated: Gated DeltaNet) delays memory decay via matrix updates, achieving 38% accuracy at $N = 1024$; Class III (Scalar Gated: GLA) transitions to 0% for $N \gtrsim 512$; and Class IV (Pure SSM: Mamba-2) reaches state capacity limits for $N \gtrsim 256$.

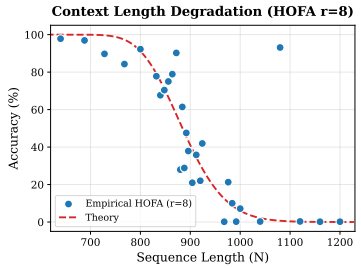
Empirical Capacity Bounds. Figure 4b evaluates the $r = 8$ model via held-out inference with a fixed lookback gap, using average per-token accuracy, the direct empirical estimate of the single retrieval event probability, modeled by Equation 11. Accuracy shows a broad downward trend consistent with the theoretical extreme value capacity model, though individual points, including a localized peak near $N = 1080$, exhibit high variance consistent with the sampling sensitivity of synthetic associative recall tasks noted in Section 4.

Feature Norm Disparity. Probing parameter norms across trained induction task checkpoints provides empirical support for the asymmetric amplification mechanism (γ). Standard MHA scatters weight magnitude across all dimensions to capture RoPE frequencies, whereas HOFA ($r = 8, 16$) exhibits a sharp "norm cliff" precisely at index $r-1$, where we observe that optimization concentrates parameter magnitude into the r outlier channels (reaching ≈ 1.85 for $r = 8$) while leaving inlier channels at baseline levels (≈ 0.4, Figure 4c).

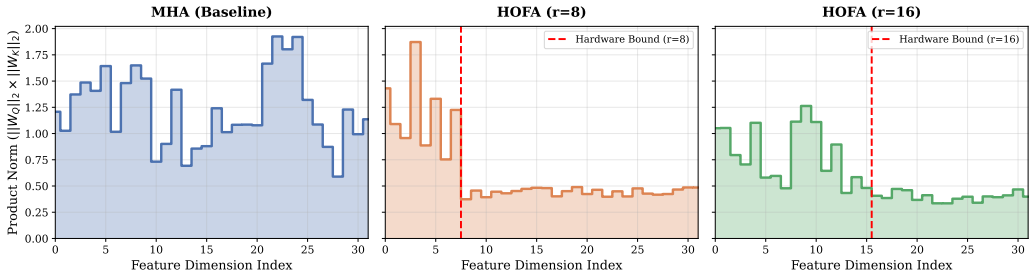
Feature-Wise Activation Norm Concentration. To motivate our choice of outlier capacity, we analyze the feature-wise activation norm decomposition [Vig and Belinkov, 2019, Clark et al., 2019] across eight production Transformer models scaling from 124M to 8B parameters (Appendix F, Figure 9). Normalizing cumulative feature activation norm by model scale shows that the cumulative feature norm concentrates within approximately the first 16 dimensions ($r \approx 16$). This observation motivates HOFA's static $r = 16$ channel allocation. The decomposition holds across two distinct architectural tracks: (i) pure MHA models (GPT-2 124M, Pythia 410M/1.4B/2.8B)



(a) Sequence Length Memory Cliff



(b) Empirical vs. Theoretical CDF Model ($r = 8$)



(c) Feature Norm Disparity Across Hidden Dimensions

Figure 4: **Mechanistic Validation & Capacity Dashboard.** (a) Associative recall accuracy vs. sequence length N across 4-layer models ($d_h = 32$). (b) Empirical degradation fitted against the theoretical capacity model. (c) Parameter magnitude (norm) across hidden dimensions in trained weights, showing sharp outlier concentration at index $r - 1$ (dashed red lines mark static hardware boundaries $r = 8, 16$).

[Xiao et al., 2023, Kovaleva et al., 2019]; and (ii) Grouped-Query Attention (GQA) [Ainslie et al., 2023] models (Llama-3.2 1B/3B, Llama-3.1 8B, Qwen2.5 7B). This provides an empirical basis for our symmetrical capacity allocation schedule, descending from $r = 64$ at boundary layers down to $r = 16$ across the middle $\sim$ 50–70% semantic core (Appendix A).

3.3 End-to-End Language Modeling

Pretraining Dynamics. To ensure robust representation learning across the dual pathways, we pretrain 70M, 125M, and 350M models on FineWeb-Edu [Penedo et al., 2024] beyond compute-optimality at 40 tokens per parameter (2$\times$ Chinchilla [Hoffmann et al., 2022]). This corresponds to 2.8B, 5B, and 14B tokens at context length $N = 1024$. As shown in Figure 5, 350M HOFA closely tracks standard MHA throughout pretraining, matching final validation perplexity at $\sim 17\%$ fewer total pretraining FLOPs. Because these pretraining runs are conducted at $N = 1024$, where feed-forward layers dominate total compute, this efficiency advantage widens at longer context lengths as attention compute scales toward the asymptotic $\frac{2d_h}{r+d_h}$ prefill speedup limit (Section 2.2, Equation 8).

Dynamic Mixing Gate Analysis. To characterize the behavior of the dynamic mixing gate α_h, we extract layer-wise distributions from the 350M HOFA model across 1,000 FineWeb-Edu validation sequences (discarding the first 5 attention-sink tokens). Contrary to binary routing expectations, token-level gate values span the full $[0, 1]$ range dynamically across sequence positions, while layer-wise mean gate values across middle and deep layers cluster tightly in a soft-blending regime ($\mu_\alpha \approx 0.38$–0.55, averaging ~ 0.45). Layer 1 exhibits a unique, low-variance concentration at $\mu_\alpha \approx 0.24$, heavily prioritizing the recurrent GLA pathway to establish initial local context (see Appendix C, Figure 7).

Effective Attention Distance: Preservation of Global Reach. To test whether confining exact Softmax to r dimensions limits global attention reach, we measure layer-wise effective attention distance across 1,000 FineWeb-Edu validation sequences. Appendix D (Figure 8) shows 350M HOFA's outlier pathway tracking standard MHA ($d_h = 128$) across all 24 layers (averaging 257.1 vs. 255.9 tokens). Conversely, the GLA inlier pathway operates at a sharp local average of 3.9 tokens. This indicates that HOFA preserves global attention reach, concentrating long-range associative matching in the outlier subspace while the recurrent linear state primarily handles immediate local context.

Language Modeling and Zero-Shot Evaluation. We benchmark HOFA against standard MHA by measuring validation perplexity, alongside zero-shot performance against established baselines across six common-sense tasks using `lm-evaluation-harness` [Gao et al., 2021]. As detailed in Table 1, 125M HOFA outperforms matched 125M MHA across 4 of 6 benchmarks while remaining competitive with academic baselines such as GPT-2 [Radford et al., 2019], Pythia-160M [Biderman et al., 2023], and Mamba-130M [Gu and Dao, 2023]. At the 350M parameter scale, HOFA outperforms matched 350M MHA across all six benchmarks, while scoring higher on ARC-E, ARC-C, and OBQA than higher-token baselines like OPT-350M [Zhang et al., 2022], Pythia-410M [Biderman et al., 2023], and Mamba-370M [Gu and Dao, 2023].

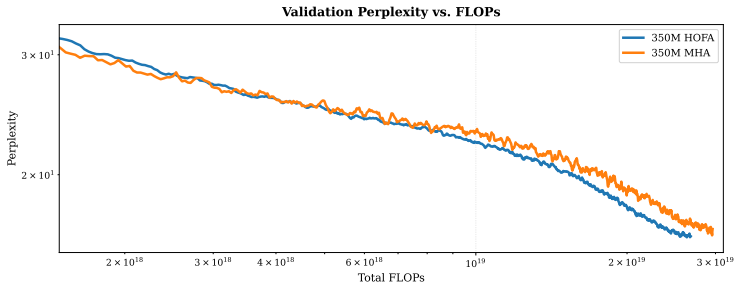


Figure 5: **350M Pretraining Dynamics.** Validation perplexity vs. pretraining FLOPs on FineWeb-Edu. 350M HOFA closely tracks standard MHA throughout training, matching final validation perplexity at $\sim 17\%$ fewer total FLOPs.

Table 1: **Validation Perplexity and Zero-Shot Common-Sense Reasoning (%).** Evaluated across six common-sense reasoning benchmarks and on held-out FineWeb-Edu validation perplexity. All models, including external baselines, were re-evaluated locally under matched evaluation settings; external baselines report (—) for perplexity due to tokenizer vocabulary and pretraining domain incomparability. **Bold** denotes overall best in scale class; underline denotes best in 1:1 controlled MHA vs. HOFA pair. Higher is better (↑) for reasoning tasks; lower is better (↓) for perplexity.

Model	Tokens	Val PPL ↓	ARC-E ↑	ARC-C ↑	PIQA ↑	Wino ↑	OBQA ↑	Hella ↑
125M MHA	5B	23.25	**46.2**	24.8	61.3	49.7	29.4	31.2
125M HOFA	5B	22.99	44.9	**26.5**	61.2	51.7	**30.2**	31.8
GPT-2 (124M)	10B	—	39.5	22.7	62.5	51.6	27.2	31.1
Pythia-160M	300B	—	39.4	23.6	61.2	52.1	26.8	30.1
Mamba-130M	300B	—	41.9	24.3	**63.2**	**52.5**	28.6	**35.2**
350M MHA	14B	15.88	50.4	26.3	66.5	51.1	32.2	41.0
350M HOFA	14B	15.87	**53.8**	**28.8**	66.8	51.5	**34.2**	42.1
OPT-350M	180B	—	40.1	23.7	64.4	52.4	28.4	36.8
Pythia-410M	300B	—	46.0	24.4	67.1	53.5	29.4	40.6
Mamba-370M	300B	—	48.2	27.9	**68.4**	**55.6**	31.2	**46.5**

Macro-Axis Structural Controls. Pretraining a 70M parameter ablation suite (2× Chinchilla-optimal budget) under parameter and FLOP parity shows that intra-head HOFA outperforms both Depth-Axis (Jamba-style) and Width-Axis (Hymba-style) hybrids on downstream reasoning, with context-aware gating contributing positively to pathway fusion. Full ablation curves and configurations are detailed in Appendix G.

4 Conclusion & Limitations

Limitations & Future Work. While HOFA establishes an effective channel-wise hybridization paradigm, several limitations should be noted. First, native pretraining is evaluated up to the 350M parameter scale on a single-node setup; scaling to billion-parameter regimes remains for future work. Second, static r-schedules are guided by post-hoc attention mass analysis of pretrained dense models (Section 3.2), leaving dynamic or learnable allocation for future study. Third, pretraining runs reflect single-seed executions due to compute limits. Finally, synthetic associative recall tasks can be prone to sampling sensitivity; we position block copying benchmarks (Appendix B) strictly as mechanistic probes, grounding our core claims in pretraining dynamics and zero-shot reasoning. Looking ahead, extending channel hybridization to composite compression schemes (Appendix H) presents promising directions for extreme long-context scaling, while deep, narrow-head HOFA architectures offer attractive design points for edge inference [Dao et al., 2023, Jiang et al., 2025].

Conclusion. We introduce HOFA, an intra-head hybrid attention layer that factorizes the hidden dimension into exact and recurrent processing streams. A dedicated r-dimensional Softmax subspace isolates and retrieves high-magnitude semantic features, while a Gated Linear Attention state processes the remaining channels to absorb background context. This decouples global associative recall from temporal state decay, recovers both pure linear and dense attention as limiting architectural cases, and enables the exact-attention pathway to preserve long-range associative retrieval while the linear pathway handles local and background context. Under matched pretraining up to 350M parameters, HOFA matches the dense MHA baseline across language modeling, commonsense reasoning, and synthetic retrieval, while reducing the decoding KV-cache footprint and improving peak throughput. Mechanistic analyses show that both pathways contribute to the hybrid state, with optimization naturally concentrating the long-range associative capacity into the Softmax outlier channels.

Reproducibility Statement. Hyperparameters and theoretical proofs are documented in Appendices A and E. Custom Triton kernels are detailed in Appendix I, with full pretraining and evaluation code available at `https://github.com/LotanGutman/OutlierNeuronAttention`.

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A Full Hyperparameter & Model Configuration Tables

Profiling was executed on one NVIDIA RTX 4060 GPU, and all pretraining runs were executed on one NVIDIA RTX 5090 GPU, requiring approximately 4 days for the 350M parameter model. Table 2 lists full pretraining hyperparameters, and Table 3 details the exact layer-by-layer heterogeneous r-schedules across model scales.

Table 2: **Full Pretraining Hyperparameters.**

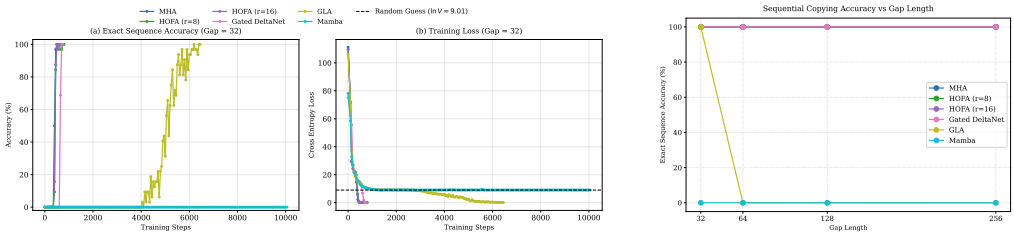
Parameter	**70M**	**125M**	**350M**
Layers (L)	8	12	24
Hidden Dim (d_{model})	512	768	1024
Heads (H)	8	6	8
Training Tokens	2.8B	5.0B	14.0B
Batch Size (sequences)	64	64	128
Sequence Length (N)	1024	1024	1024
Optimizer	AdamW	AdamW	AdamW
Weight Decay	0.1	0.1	0.1
Adam β_1, β_2	0.9, 0.95	0.9, 0.95	0.9, 0.95
Warmup Steps	2000	2000	2000
Precision	bfloat16	bfloat16	bfloat16

Table 3: **Heterogeneous Outlier Routing Schedules (r-schedules) Across Model Scales.**

Model Scale	**Layers (L)**	**Layer-by-Layer Outlier Dimension Schedule (r)**
70M	8	$[64, 32, 16, 16, 16, 16, 32, 64]$
125M	12	$[64, 32, 16, 16, 16, 16, 16, 16, 16, 16, 32, 64]$
350M	24	$[64, 32, 32, \underbrace{16, \ldots, 16}_{\times 18}, 32, 32, 64]$

B Extended Multi-Token Block Copying Benchmarks

To isolate gap driven temporal state decay from pattern capacity limits, we evaluate 4-layer models ($L = 4, d_{\text{model}} = 128, H = 4, d_h = 32$, vocab $= 8192$) on multi-token block copying across noise gaps $G \in \{32, 64, 128, 256\}$ at pattern length $P = 2$. As shown in Figure 6a, Class I (MHA, HOFA) and Class II (Gated DeltaNet) converge within ~ 500 steps at $G = 32$, whereas Class III (GLA) requires ~ 5000 steps. Across longer gaps $G \geq 64$ (Figure 6b), GLA collapses to 0% accuracy from recurrent state collisions, while HOFA reliably maintains exact recall up to $G = 256$.



(a) Convergence ($G = 32$). (b) Accuracy vs. noise gap G.

Figure 6: **Multi-Token Block Copying Benchmarks ($P = 2$). (a)** Class I (MHA, HOFA) and Class II (Gated DeltaNet) converge in ~ 500 steps at $G = 32$, while Class III (GLA) requires ~ 5000 steps and Class IV (Mamba-2) flatlines. **(b)** Exact recall accuracy vs. noise gap G; by $G \geq 64$, GLA collapses to 0%, while HOFA preserves retrieval up to $G = 256$.

C Dynamic Mixing Gate Distribution

To characterize dynamic mixing gate behavior, we extract layer-wise distributions from the 350M HOFA model ($L = 24$) across 1,000 validation sequences. As shown in Figure 7, token-level gate values span the full $[0, 1]$ range, demonstrating dynamic, context-dependent blending rather than hard binary routing. Across middle and deep layers, mean gate values cluster in a soft-blending regime. Layer 1 displays an anomalous low-variance distribution, reflecting a strong initial reliance on the recurrent GLA state for local context aggregation.

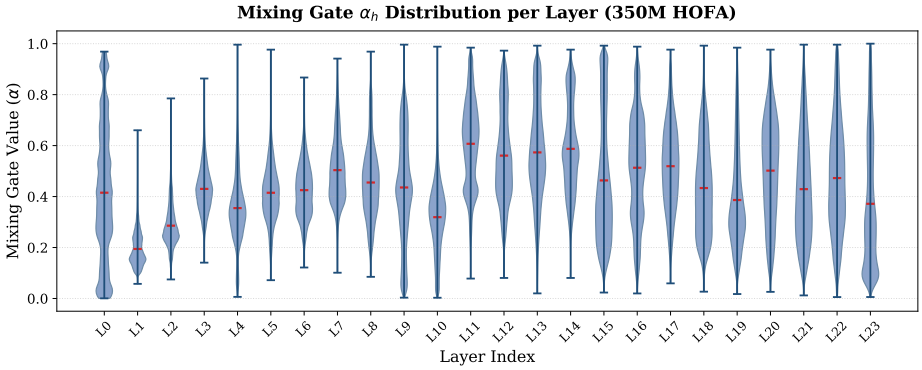


Figure 7: **Mixing Gate α_h Distribution per Layer (350M HOFA).** Layer-wise empirical distribution of the dynamic mixing gate across 24 layers. Token-level gate values span $[0, 1]$ dynamically, while layer-wise means across middle and deep layers cluster in a soft-blending regime ($\mu_\alpha \approx 0.38$–0.55, horizontal ticks). Layer 1 exhibits anomalous, low-variance reliance on the inlier GLA pathway ($\mu_\alpha \approx 0.24$).

D Effective Attention Distance Analysis

To evaluate whether HOFA maintains long range context dependency comparable to full Softmax attention, we compute the layer wise effective attention distance d_l across 1,000 validation sequences from FineWeb-Edu following standard attention distance methodologies. For standard MHA and HOFA's exact outlier pathway, d_l is defined as:

$$d_l^{\text{outlier}} = \frac{1}{N-5} \sum_{i=6}^{N} \sum_{j \le i} p_{ij}^{(l)} (i-j), \quad \text{where } p_{ij}^{(l)} = \text{softmax}_{j \le i} \left(\frac{1}{\sqrt{r}} \widetilde{Q}_{\mathcal{O},i}^{(l)} (\widetilde{K}_{\mathcal{O},j}^{(l)})^T + M_{ij} \right) \quad (12)$$

discarding the first 5 attention-sink tokens [Xiao et al., 2023]. Because the unrolled linear attention state yields unbounded and potentially negative values, we introduce a magnitude-normalized pseudo-probability $P_{ij}^{(l)}$ to enable direct distance comparisons with Softmax distributions. This is derived from the inlier feature content $A_{ij}^{(l)}$ and the cumulative gate decay matrix $D_{ij}^{(l)}$:

$$A_{ij}^{(l)} = \widetilde{Q}_{\mathcal{I},i}^{(l)} (\widetilde{K}_{\mathcal{I},j}^{(l)})^T, \qquad D_{ij}^{(l)} = \prod_{s=j+1}^{i-1} \sigma(-g_s^{(l)}) \quad (j < i), \quad (13)$$

$$P_{ij}^{(l)} = \frac{|D_{ij}^{(l)} A_{ij}^{(l)}|}{\sum_{k<i} |D_{ik}^{(l)} A_{ik}^{(l)}|}, \qquad d_l^{\text{inlier}} = \frac{1}{N-5} \sum_{i=6}^{N} \sum_{j<i} P_{ij}^{(l)} (i-j). \quad (14)$$

Figure 8 shows the 350M HOFA outlier pathway tracking standard MHA across all 24 depth layers (averaging 257.1 vs. 255.9 tokens). Conversely, the GLA pathway operates at a sharp local average of 3.9 tokens. This provides empirical evidence that long-range associative matching is concentrated in the outlier subspace, while the recurrent linear state primarily captures immediate local context.

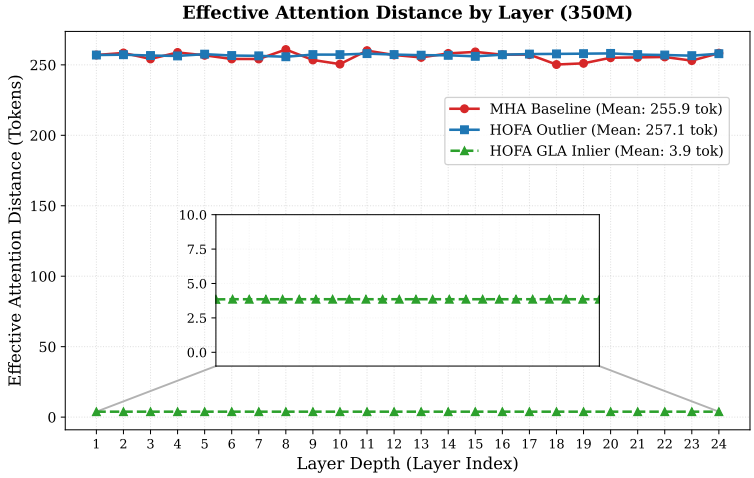


Figure 8: **Layer-Wise Effective Attention Distance (350M Model).** Comparison of standard MHA ($d_h = 128$), HOFA Outlier pathway, and HOFA GLA Inlier pathway across all 24 Transformer layers. The outlier pathway preserves MHA's full global attention reach (257.1 vs. 255.9 tokens), while the GLA state absorbs local contextual background (3.9 tokens).

E Extended Mathematical Proofs & Derivations

The Welch Bound and the Scaling Paradox. Packing N key vectors into a restricted r-dimensional routing subspace creates inevitable geometric interference. By Welch's theorem [Welch, 1974], the aggregate off-diagonal squared inner products of N unit vectors satisfy:

$$\sum_{i=1}^{N}\sum_{j\neq i}^{N} |\langle x_i, x_j\rangle|^2 \geq \frac{N^2}{r} - N = \frac{N(N-r)}{r} \tag{15}$$

Whereas classical coding theory often considers pairwise maximum correlation [Welch, 1974], associative retrieval in attention aggregates interference across all $N-1$ competing tokens simultaneously. The expected per-vector interference power seen by a query vector is:

$$I = \frac{1}{N}\sum_{i=1}^{N}\sum_{j\neq i}^{N} |\langle x_i, x_j\rangle|^2 \geq \frac{N-r}{r} \tag{16}$$

which scales linearly with sequence length N. Assuming unit signal alignment for the target key ($\langle q, k_0\rangle^2 = 1$), Equation 16 dictates that the naive Signal-to-Noise Ratio is bounded from above by an achievable capacity ceiling:

$$\text{SNR}_{\text{naive}} = \frac{1}{I} \leq \frac{r}{N-r} \tag{17}$$

To model structured representations in multi-head attention, we introduce the semantic subspace partitioning hypothesis: keys are distributed across d_h approximately orthogonal feature directions, concentrating query competition into an active subspace pool of cN tokens with distractor density $c = 1/d_h$. Within this shared semantic subspace, the naive SNR ceiling becomes:

$$\text{SNR}_{\text{naive}} \leq \frac{r}{cN-r} \tag{18}$$

While continuous interference variance scales with $(cN - r)$, discrete target retrieval faces $n = cN - 1 = N/d_h - 1$ competing distractors. This establishes an apparent scaling paradox: if the network attempts to boost signal separation by uniformly scaling projection weights by an isotropic magnitude factor S ($q \leftarrow S\hat{q}, k \leftarrow S\hat{k}$), target signal power scales as S^2, but distractor noise variance scales proportionally as $S^2 I \geq S^2 \frac{cN-r}{r}$. The factor S^2 cancels algebraically in the SNR ratio, leaving the naive SNR ceiling strictly invariant to uniform magnitude scaling.

Asymmetric Gradient Gain. We model HOFA's feature crystallization as an asymmetric optimization process. Cross-entropy loss rewards increasing the true target logit z_0 relative to distractors z_i, applying an asymmetric, anisotropic gradient pressure. The optimizer selectively amplifies discriminative subspace directions responsible for pattern matching, while ambient distractor directions receive diffuse gradients under the Softmax partition function. We define $\gamma \gg 1$ as this loss-driven asymmetric gain factor, yielding the effective Signal-to-Noise Ratio:

$$\text{SNR}_{\text{eff}} \approx \gamma \frac{r}{cN - r}, \quad \mu = \sqrt{\text{SNR}_{\text{eff}}} \approx \sqrt{\gamma \frac{r}{cN - r}} \tag{19}$$

This anisotropic gain is empirically manifested by the sharp "norm cliff" in outlier feature projections (Section 3.2, Figure 4c), where optimization concentrates high-magnitude associative matching into the r outlier channels.

M-ary Hypothesis Testing & The Gumbel Threshold. For successful retrieval, the target logit $z_0 \sim \mathcal{N}(\mu, 1)$ must exceed the maximum distractor logit $M_n = \max_{i=1}^{n} z_i$ across $n = N/d_h - 1$ competing keys. Under the Fisher-Tippett-Gnedenko theorem [Fisher and Tippett, 1928], the maximum of n Gaussian distractors converges to a Gumbel distribution with leading-order threshold $b_n \approx \sqrt{2 \ln n}$. The retrieval probability is:

$$P(\text{correct}) = P(z_0 > M_n) \approx \Phi\left(\frac{\mu - b_n}{\sigma}\right) \tag{20}$$

Here, σ is an empirical calibration parameter. While standard extreme value theory assumes independent unit-variance distractors, real transformer representations exhibit feature correlations and non-unit variance; σ calibrates the effective transition width without altering the underlying Gumbel scaling trajectory.

F Attention Mass Distribution Across Model Scales

To quantify how attention weight concentrates across latent feature dimensions, we compute the cumulative Probability Mass Decomposition across query heads:

$$\mathcal{C}(r) = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \sum_{k=1}^{r} p_q^{(k)} \tag{21}$$

where $p_q^{(k)}$ denotes the sorted attention weight allocated to feature dimension k.

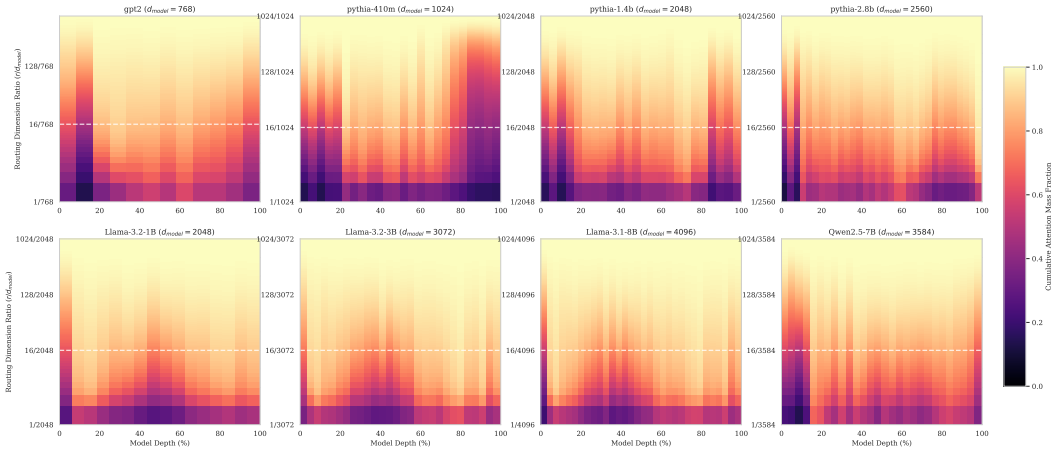


Figure 9: **Feature-Wise Activation Norm Decomposition Across 8 Production Models.** Empirical motivation for HOFA's static r-schedule. Top row: Pure MHA track (GPT-2 124M to Pythia-2.8B). Bottom row: GQA spatial compression track (Llama-3.2 1B/3B, Llama-3.1-8B, Qwen2.5-7B). The routing dimension $r \approx 16$ (dashed white line) captures a similar fraction of cumulative attention mass as model size scales up to 8B parameters.

G Extended Ablation Suite & Macro-Axis Controls

To evaluate the intra-head channel-wise hybridization axis against macro-structural alternatives, we pretrain a 70M parameter ablation suite ($L = 8, d_{\text{model}} = 512, H = 8, d_h = 64$) on 2.8B tokens ($2\times$ Chinchilla-optimal budget, 40 tokens/parameter). Zero-shot reasoning benchmarks (Table 4) and validation perplexity curves (Figure 10) reflect parameter and FLOP parity across all 70M variants.

Table 4: **70M Parameter Scale Validation Perplexity and Zero-Shot Common-Sense Reasoning (%).** Evaluated across six common-sense reasoning benchmarks and on held-out FineWeb-Edu validation perplexity. All models are pretrained under parameter and compute parity ($2\times$ Chinchilla-optimal, 2.8B tokens). **Bold** denotes best performance among hybrid configurations. Higher is better (↑) for reasoning tasks; lower is better (↓) for perplexity.

Model (70M Scale)	Val PPL ↓	ARC-E ↑	ARC-C ↑	PIQA ↑	Wino ↑	OBQA ↑	Hella ↑
MHA Baseline	31.37	**41.2**	**25.4**	57.7	50.6	28.6	28.2
Depth-Axis Hybrid (GLA + MHA)	31.54	40.8	22.7	59.1	48.8	28.8	28.0
Width-Axis Hybrid (GLA + MHA)	31.31	40.4	23.3	59.0	50.9	30.2	27.7
HOFA (Flat $r = 32$ + Dynamic Gate)	30.96	41.1	24.1	57.9	50.8	30.6	27.8
Canonical HOFA (U-Shaped + Dynamic Gate)	**30.65**	40.6	23.7	**60.2**	**51.1**	**30.8**	**28.5**
Static-Blend HOFA (− Dynamic Gate)	30.85	39.7	24.6	58.8	**51.1**	30.6	28.2

Capacity Matching (Average $r = 32$). To establish a fair comparison, all macro axis baselines are parameterized to match Canonical HOFA's total exact attention capacity. Canonical HOFA employs a heterogeneous, boundary heavy routing schedule ($[64, 32, 16, 16, 16, 16, 32, 64]$; Table 3), yielding a cumulative budget of 256 exact attention channels across 8 layers (an average of $r = 32$ per layer). Consequently, the Uniform HOFA control fixes $r = 32$ across all layers, while the Depth-Axis and Width-Axis hybrids allocate exactly 50% of their total capacity (256 out of 512 total channels) to exact MHA, ensuring $1:1$ parameter and FLOP equivalence.

Controlled Depth-Axis Baseline. The Depth-Axis Hybrid (Jamba-style) alternates 4 pure MHA layers ($d_h = 64$) and 4 pure GLA layers ($d_h = 64$) using a symmetric $[0, 64, 64, 0, 0, 64, 64, 0]$ schedule. We deliberately omit full attention from the boundary layers (Layers 0 and 7) so that the depth baseline does not implicitly benefit from HOFA's boundary heavy capacity thesis (Section 3.2, Figure 9). Although standard depth-hybrids employ much sparser exact attention ratios (e.g., 1:7), we enforce a strict 1:1 ratio (50% capacity) for this baseline. This ensures that downstream performance differences isolate the architectural routing geometry, rather than variations in total exact attention capacity.

Micro-Architecture Control (GLA vs. Mamba). Across all hybrid baselines (Depth-Axis and Width-Axis), the linear pathway is implemented using our exact Gated Linear Attention (GLA) kernel. Holding the linear micro-architecture constant isolates the structural hybridization axis (layer-wise vs. head-wise vs. channel-wise) as the sole independent variable.

Static vs. Dynamic Blending. In addition to macro axis baselines, we evaluate a Static-Blend HOFA variant where the dynamic token-level mixing gate α_h is replaced by a fixed, layer-wise scalar, isolating the contribution of context-aware dynamic gating.

Ablation Results & Analysis. Results from our ablation suite, detailed in Table 4 and Figure 10, demonstrate the advantages of channel-wise hybridization. While the Homogeneous ($r = 32$) HOFA baseline performs competitively, upgrading to the heterogeneous, boundary-heavy U-shaped r-schedule yields higher downstream accuracy. This aligns with our attention mass decomposition (Figure 9), confirming that allocating exact capacity to boundary layers outperforms a flat distribution. Canonical HOFA outperforms both Depth-Axis (Jamba-style) and Width-Axis (Hymba-style) hybrids across 4 of 6 downstream reasoning benchmarks (including PIQA, WinoGrande, OBQA, and HellaSwag), while optimizing faster to reach target perplexities earlier under matched compute (Figure 10). Finally, removing the dynamic mixing gate (Static-Blend HOFA) degrades accuracy, indicating that dynamic per-token routing projection benefits exact-linear pathway fusion over a static scalar.

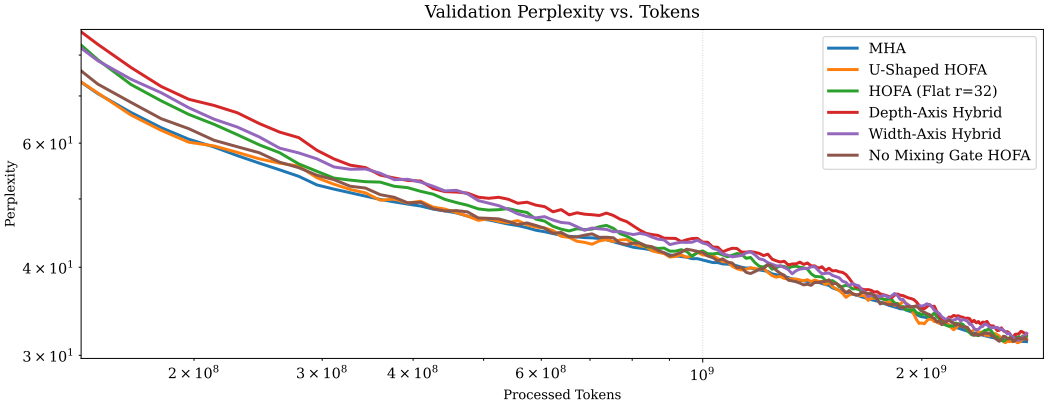


Figure 10: **Validation Perplexity Curves for the 70M Parameter Ablation Suite.** Models are trained on 2.8B tokens (2× Chinchilla-optimal). All hybrid baselines are restricted to an identical 50% exact attention capacity budget. U-Shaped HOFA and the No Mixing Gate variant track the pure MHA baseline with similar optimization trajectories, whereas the Depth-Axis and Width-Axis hybrids exhibit slower initial optimization during the first 10^9 tokens. (A rolling smoothing window is applied to all curves to filter batch-wise variance).

H Orthogonal Architectural Extensions: Composite GQA and MLA Formulations

Multi-Head Latent Attention (MLA) [DeepSeek-AI, 2024] and Grouped-Query Attention (GQA) [Ainslie et al., 2023] are prominent strategies for KV-cache memory reduction, reducing memory footprint via low-rank projection or spatial key sharing. These mechanisms operate along structural dimensions orthogonal to HOFA's intra-head channel decomposition. Below, we propose conceptual composite formulations showing how HOFA can be combined with both mechanisms.

Composite HOFA–GQA. Grouped-Query Attention (GQA) achieves spatial compression by forcing G Query heads to share single Key and Value heads. In contrast, HOFA provides temporal compression by offloading non-outlier channels to a recurrent GLA state. In a composite HOFA–GQA layer, G Query heads share a single r-dimensional Softmax Key head alongside a shared $(d_h - r)$-dimensional GLA inlier state, combining spatial head-sharing with temporal channel compression. A practical consideration is that aggressive spatial head sharing ($G \geq 8$) may constrain key diversity across heads, requiring appropriate selection of r.

Composite HOFA–MLA. Multi-Head Latent Attention (MLA) provides low-rank latent compression by projecting key-value states into compact latent spaces. Specifically, MLA requires a decoupled positional key path because low-rank compression and rotary embeddings do not commute; since HOFA already confines RoPE exclusively to the r-dimensional outlier subspace, this decoupling is inherently satisfied without additional positional projection machinery. Low-rank latent compression can therefore be applied directly to the unpartitioned Value stream V alone by caching a single shared low-rank latent vector $c_t^V \in \mathbb{R}^{d_c}$ per token (where $d_c < Hd_h$), reconstructing per-head value states at read time. This reduces the per-token cache footprint from $H(r + d_h)$ to $Hr + d_c$, showing that intra-head channel factorization is structurally compatible with latent compression. Retaining uncompressed outlier keys ($K_{\mathcal{O}}$) separates retrieval precision from value-stream compression, while decompressing value vectors introduces additional reconstruction operations during decoding.

We emphasize that these composite formulations serve as conceptual blueprints that suggest HOFA may be structurally compatible with orthogonal KV-compression mechanisms. We leave empirical training and hardware benchmarking of these composite configurations to future work.

I Implementation & Triton Kernel Engineering

To optimize the exact outlier pathway under FlashAttention execution logic, we implement Causal Loop Peeling [Dao, 2023] in a custom Triton kernel; complete kernel implementations and evaluation suites are provided in the linked anonymized repository. Below is a streamlined structural excerpt of the fused forward kernel (`_fwd_kernel`) illustrating the two-phase loop peeling execution (unmasked block-sparse loop below diagonal and masked diagonal block):

```
@triton.jit
def _fwd_kernel(
    Q, K, V, Out, mix_g_ptr, sm_scale,
    stride_qz, stride_qh, stride_mix_g_n, ....,
    Z, H, N_CTX, r,
    BLOCK_M: tl.constexpr, BLOCK_N: tl.constexpr
):
    start_m = tl.program_id(0)
    off_hz = tl.program_id(1)

    # Unmasked blocks below diagonal
    for start_n in range(0, start_m * BLOCK_M, BLOCK_N):
        k = tl.load(k_ptrs, mask=mask_qk[:, None], other=0.0)
        qk = tl.dot(q, k, allow_tf32=True) * sm_scale
        m_ij = tl.maximum(m_i, tl.max(qk, 1))
        p = tl.exp(qk - m_ij[:, None])
        l_i = l_i * tl.exp(m_i - m_ij) + tl.sum(p, 1)
        acc = acc * tl.exp(m_i - m_ij)[:, None] + tl.dot(p.to(v.dtype), v)
        m_i = m_ij

    # Masked diagonal block
    for start_n in range(start_m * BLOCK_M, (start_m + 1) * BLOCK_M, BLOCK_N
        ):
        k = tl.load(k_ptrs, mask=mask_k, other=0.0)
        qk = tl.dot(q, k, allow_tf32=True) * sm_scale
        qk = tl.where(offs_m[:, None] >= offs_n_curr[None, :], qk, float("-
            inf"))
        m_ij = tl.maximum(m_i, tl.max(qk, 1))
        p = tl.exp(qk - m_ij[:, None])
        acc = acc * tl.exp(m_i - m_ij)[:, None] + tl.dot(p.to(v.dtype), v)
```